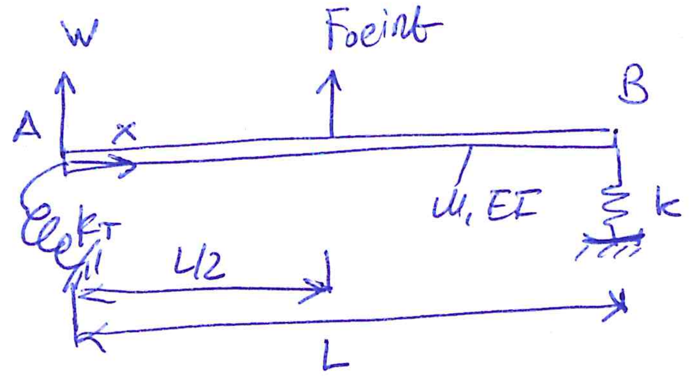
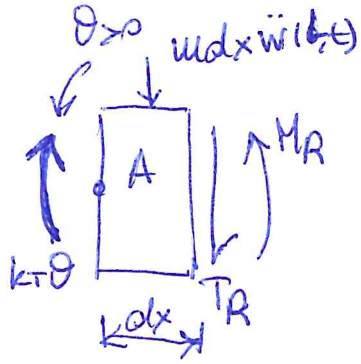
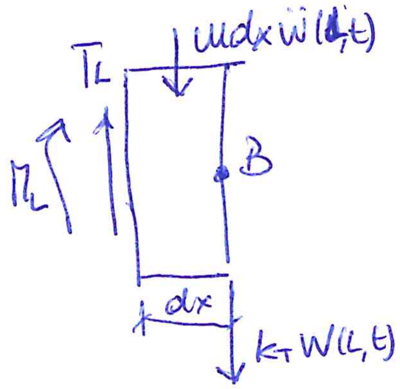
Written Test August 31st 2022

PROBLEM 2







1) $k_T\theta(0,t) - M_R + \cancel{T_R dx} + \cancel{m dx \ddot{w}(0,t)\frac{dx}{2}} = 0$

$k_T\theta(0,t) - M(0,t) = 0$

2) $T_R + \cancel{m dx \ddot{w}(0,t)} = 0$

$T(0,t) = 0$

3) $M_L + \cancel{T_L dx} - \cancel{m dx \ddot{w}(L,t)\frac{dx}{2}} = 0$

$M(L,t) = 0$

4) $T_L - \cancel{m dx \ddot{w}(L,t)} - k_T w(L,t) = 0$

$T(L,t) - k_T w(L,t) = 0$

$$
\begin{cases}
k_T\gamma(B+D) - \gamma^2 EI(-A+C) = 0 \\
\cancel{\gamma^3 EI}(-B+D) = 0 \\
\cancel{\gamma^2 EI}(-A\cos\gamma L - B\sin\gamma L + C\cosh\gamma L + D\sinh\gamma L) = 0 \\
\gamma^3 EI(A\sin\gamma L - B\cos\gamma L + C\sinh\gamma L + D\cosh\gamma L) - k(A\cos\gamma L + B\sin\gamma L + C\cosh\gamma L + D\sinh\gamma L) = 0
\end{cases}
$$

$$
H = \begin{bmatrix}
\frac{\gamma EI}{k_T} & 1 & -\frac{\gamma EI}{k_T} & 1 \\
0 & -1 & 0 & 1 \\
-\cos\gamma L & -\sin\gamma L & \cosh\gamma L & \sinh\gamma L \\
C_1\sin\gamma L + -\cos\gamma L & -C_1\cos\gamma L + -\sin\gamma L & C_1\sinh\gamma L + -\cosh\gamma L & C_1\cosh\gamma L + -\sinh\gamma L
\end{bmatrix}
$$

$\underline{x} = \{A \quad B \quad C \quad D\}$

$m_{mi} = m_{mi,\,beams}$

$k_{mi} = k_{mi,\,beams} + k_T\phi_i'^2(0) + k\phi_i^2(L)$

ϕ_i = i-th row of the output matrix of the "modeshapes" function